Formulating your world as MDP

Let's call the state space by "S", Action space by "A" , Rewards by "R" and Transition probability by "T" :

S={ S0- Wake up , S1- Go for bath , S2- Sleep for 30 minutes more }

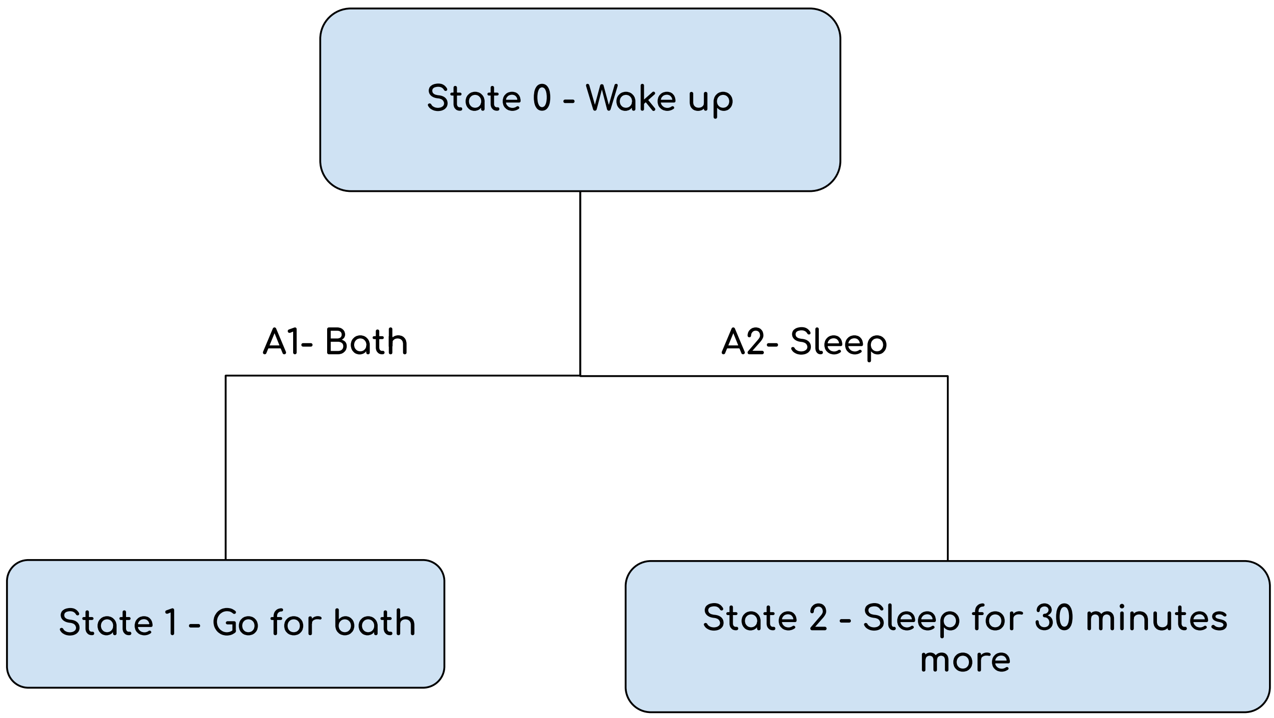
A={ A1- bath , A2- sleep }

Taking A1 action has an 80% chance of transitioning to State 1 (Go for bath) and a 20% chance of transitioning to State 2 (Sleep for 30 minutes more).

Taking A2 action has a 10% chance of transitioning to State 1 (Go for bath) and a 90% chance of transitioning to State 2 (Sleep for 30 minutes more).

State 0 (Initial state)
States 1 & 2 (The Terminals)

When we transition from state 0 (wake up) to state 1 (go for bath) we get a reward of **+10**. Otherwise when we transition from state 0 to state 2 (Sleep for 30 minutes more) we get a reward of **+5**.



$$V_{k+1}(s) = \max_a \sum_{s'} P(s'|s,a)\,[R(s') + \gamma V_k(s')]$$

Calculating value iteration for exactly 2 steps ,assuming gamma factor (discount factor) =0.9

Initially V0(S)=0 for all states.

We know that the value function for state 1&2 will remain zero because they are the terminal states

If action = A1 (study):
V1(S0) = 0.8(10 + 0.9(0)) + 0.2(5 + 0.9(0))=9
V2(S0)= 0.8(10 + 0.9(0)) + 0.2(5 + 0.9(0))=9

If action = A2 (sleep):
V1(S0) = 0.1(10 + 0.9(0)) + 0.9(5 + 0.9(0))=5.5
V2(S0) = 0.1(10 + 0.9(0)) + 0.9(5 + 0.9(0))=5.5

So if our policy maps the initial state to terminal state by action A1
Then our value function after 2 iteration is
V2(S0)=9 , V2(S1)=0 , V2(S2)=0.

Else if the policy maps the initial state to terminal state by action A2
Then our value function after 2 iteration is
V2(S0)=5.5 , V2(S1)=0 , V2(S2)=0.